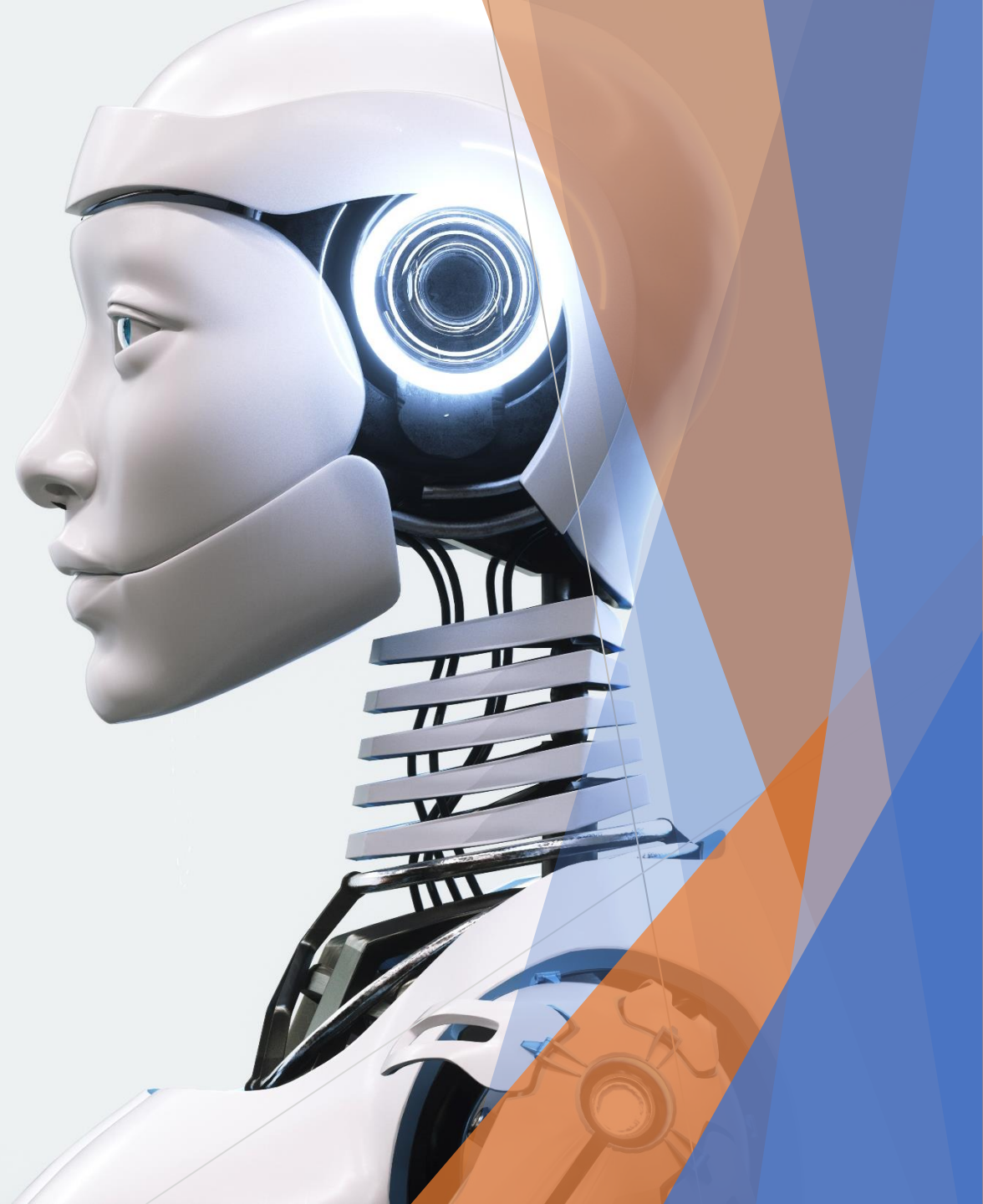




Amazon Bedrock

With Agentic AI



Amazon Bedrock

The easiest way to build
and scale generative AI
applications with
foundation models (FMs)

Definition: Generative AI

- ▶ Generative artificial intelligence (also generative AI or GenAI) is artificial intelligence capable of generating text, images, or other media, using generative models.
- ▶ Generative AI models learn the patterns and structure of their input training data and then generate new data that has similar characteristics.

-Wikipedia

Foundation Model:

A large machine learning (ML) model trained on a vast quantity of data

- ▶ A foundation model is a large machine learning model, trained on a vast quantity of data at scale such that it can be adapted to a wide range of downstream tasks.
- ▶ An FM uses learned patterns and relationships to predict the next item in a sequence.
- ▶ Foundation models, even though are pre-trained, can continue to learn from data inputs or prompts during inference. This means that you can develop comprehensive outputs through carefully curated prompts.
- ▶ Tasks that FMs can perform include language processing, visual comprehension, code generation, and human-centered engagement.
- ▶ Inference is the process of running live data through a trained AI model to make a prediction or solve a task.

Amazon Bedrock

- ▶ Amazon Bedrock is a fully managed service that makes FMs(foundation models) from leading AI startups and Amazon available, via an API.
- ▶ You can quickly experiment with a variety of FMs in the playground and use a single API for inference regardless of the models you choose.
- ▶ Since Bedrock is serverless, you can get started quickly, privately customize FMs with your own data, and easily integrate and deploy them into your applications.

Choose between high performing FMs



Jurassic-2

Instruction-following LLMs for any language task including question answering, summarization, text generation, and more



Command

Transform your products with foundational large language models built for enterprises



Amazon Titan

Text summarization, generation, classification, open-ended Q&A, information extraction, embeddings and search



Claude

LLM for conversations, question answering, and workflow automation based on research into training honest and responsible AI systems



Stable Diffusion

Generation of unique, realistic, high-quality images, art, logos, and designs



Llama 2

Meta is looking to unlock the power of large language models. Our latest version of Llama is now accessible to individuals, creators, researchers and businesses of all sizes so that they can experiment, innovate and scale their ideas responsibly

Benefits of Amazon Bedrock

- ▶ Choice of leading foundation models
- ▶ Easy model customization with your data
- ▶ Fully managed agents that dynamically invoke APIs to execute tasks

Amazon Bedrock Use Cases

- ▶ Text generation
- ▶ Chatbots
- ▶ Search
- ▶ Text summarization
- ▶ Image generation
- ▶ Personalization



Agentic AI

Amazon Bedrock Agents

A cutting-edge feature from AWS that lets you build powerful generative AI applications capable of automating complex, multi-step business tasks

Amazon Bedrock Agents

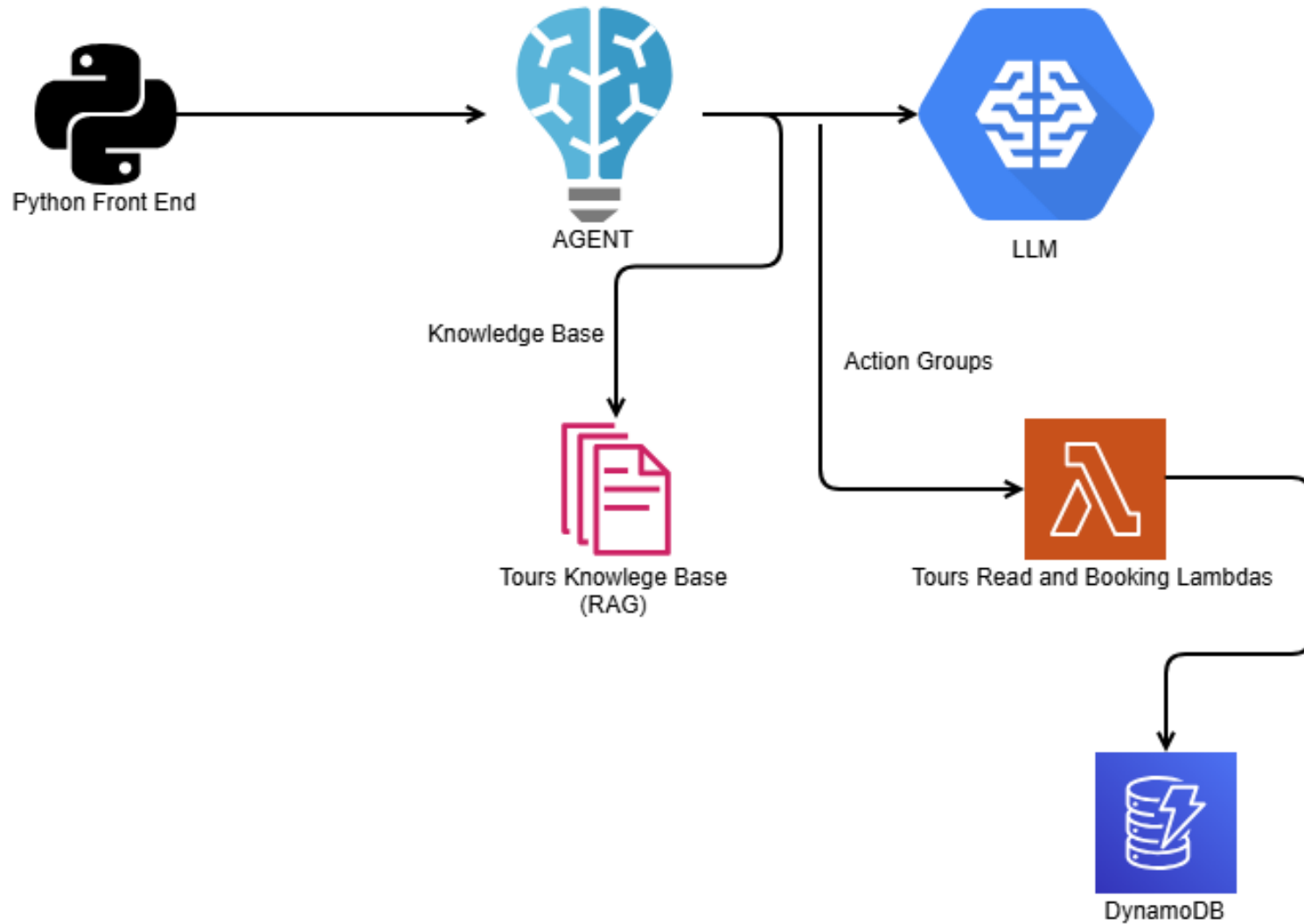
These agents use advanced foundation models to:

- 1) Understand user requests
- 2) Break them down into actionable steps
- 3) Interact with your company's systems, APIs, and data sources to get the job done—all through natural language conversations

Amazon Bedrock Agent Components

- ▶ Action Groups
- ▶ Knowledge Bases
- ▶ Memory Retention

Agentic AI Hands On



Amazon Bedrock Action Groups

Enable Bedrock agents to perform real-world tasks by defining a set of callable actions

These actions extend the agent's capabilities beyond natural language processing, allowing it to interact with external systems, trigger operations, and complete multi-step workflows

Amazon Bedrock Knowledge Bases

Enables organizations to integrate their private, proprietary data into generative AI applications

This integration allows foundation models and agents to access up-to-date, domain-specific information

Retrieval Augmented Generation (RAG)

Enhances large language models (LLMs) by allowing them to *retrieve and incorporate* up-to-date, domain-specific information from external sources before generating a response